

Force and machines

Have you ever wondered why you are more likely to slip on a wet floor than a dry one? Why there are more motor bike accidents after rainfall? Why cars skid easily on an oily road than a dry one? Why is it difficult to pull or push object on rough surfaces? What helps us to walk on different surfaces?

Look at the following figure:



Figure 6.1: Man moving a box on rough surface

This man is having difficulty in moving the large box on the road. Why is it difficult to move the box? A runner slipped on a banana peel. Why did the runner fall down? How did the banana peel contribute to the accident?



Figure 6.2 A runner slipping on banana peeling.

All the students will be able to:

- ✓ Define friction.
- ✓ Describe causes of friction.
- ✓ Explain the advantages and disadvantages of friction.
- ✓ Suggest methods to reduce friction.
- ✓ Explain air resistance.
- ✓ Identify the ways to reduce air resistance.
- ✓ Explain gravity as a force.
- ✓ Differentiate between mass and weight.
- ✓ Differentiate between balanced and unbalanced forces.
- ✓ Describe the effects of balanced and unbalanced forces on the motion of an object.
- ✓ Explain inertia.
- ✓ Describe simple machines (wedge, inclined plane and kinds of levers)
- ✓ Describe the use of wedge, inclined plane and levers in daily life.

In this chapter, you will learn about

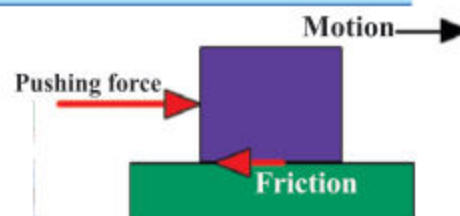
- Friction.
- Advantages and disadvantages of friction.
- Methods to reduce friction.
- Gravitational force.
- Mass and weight.
- Balanced and unbalanced forces.
- Inertia.
- Simple machine (wedge and inclined plane).
- Lever.
- Kinds of lever (scissor, hammer, plier, wheel barrow, tweezers, tongs, etc.)
- Use of lever in daily life.

Friction

- 1 Define friction.
- 1 Describe causes of friction.

Friction is a **force** that stops things from moving easily.

When two surfaces move or slide over each other, they experience **frictional force**.



This force always acts in **opposite direction** and slows down moving objects.

In the illustration fig 6.2, both the runners are wearing sport shoes to help them run easily. The shoes provide a friction between the two surfaces. The banana peeling reduces this friction and the runner falls down. Without frictional forces, a moving object may continue to move or skid on surfaces for a longer period.

Figure 6.3:
Magnified
image of
two surfaces
in contact



The surfaces that look smooth have microscopic ridges and grooves which interlock with each other and slow down the movement. That's why it is difficult

to move the large box on the road. The rougher the road, the more friction will be between the box and the road.

Rough surfaces provide more friction. Smooth surfaces provide least friction.

Have you ever pondered about the friction of your running shoes?



Why are there ridges and grooves on the soles of the shoes?

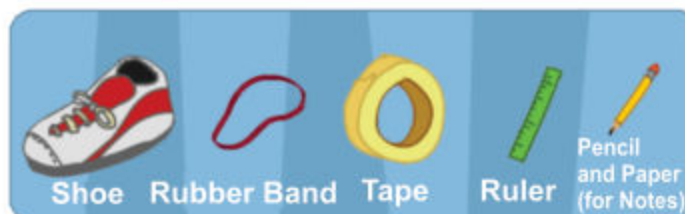
Figure 6.4: A pair of shoes with ridges and grooves



Activity 1: Investigating surfaces that cause more or less friction

What I need:

- a shoe
- a large rubber band
- a tape
- a big ruler
- a pencil and paper
- a large piece of sandpaper
- a doormat
- two clear plastic pieces



[Note : doormat, sandpaper and plastic should be of equal sizes.]

What I do :

1. Cut the rubber band and measure its length without stretching and record it in the table given below.
2. Tie the rubber band from the shoe lace hole.
3. Stick doormat, sand paper, plastic pieces on the table.
4. Spread some oil on one of the plastic piece.
5. Keep the shoe on one end of the doormat and drag it till the other end.
6. Measure the length of the rubber band in the stretched position and record it in the table.
7. Repeat steps 5 and 6 with three other surfaces that you have prepared.



What I observed:

Initial length of the rubber band	
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S. No:	Surfaces	Length of the rubber band
1.	Door mat	
2.	Sand paper	
3.	Clear plastic	
4.	Oil	

Activity questions:

1. Which surface provides the least friction? Why?
2. Which surface provides the most friction? Why?
3. What is the effect of different surfaces on the length of the rubber band?
4. On which surface the rubber band stretches the most?
5. What can you conclude about the above activity?

What I concluded:

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Advantages and disadvantages of friction

- 1. Explain the advantages and disadvantages of friction.
- 2. Think of the ways to reduce friction.

Can you easily walk on a highly polished floor with smooth soles? Can you comfortably stand on a wet soapy floor? Definitely not. In both cases we have reduced **friction**; however sometimes friction can be useful, and at other times we need to reduce it. let us study the advantages and disadvantages of friction.

Advantages of friction

1. We need friction between our shoes and the floor to walk easily.
2. Car brakes work due to the force of friction.
3. Friction also prevents knots from coming undone.
4. Nuts, bolts and nails remain fixed due to the force of friction.

Look at the pictures. How is friction involved in these processes?



In winters, why do we rub our hands together?

Friction produces heat.

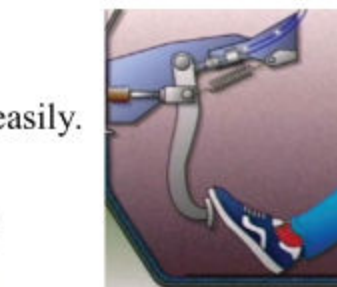
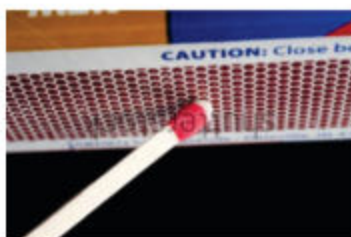


Figure 6.5: Applying brakes in car



Figure 6.6 Nut, bolts and knots



Why do we rub a match stick to burn it?

Disadvantages of friction

1. In overcoming the friction, a lot of energy is converted to non-usable form of energy and lost in the surrounding environment.
2. Friction causes wear and tear of the moving parts.
3. Due to friction, speed of automobiles cannot be increased beyond a certain limit.

Ways to reduce friction

1. By making the surfaces smooth.
2. By decreasing the area of contact of the surfaces.
3. By using oils and lubricants between the moving parts of machines.
4. By polishing the surfaces so that they can slide easily.
5. By using ball bearings because rolling objects provide less friction than the sliding ones.

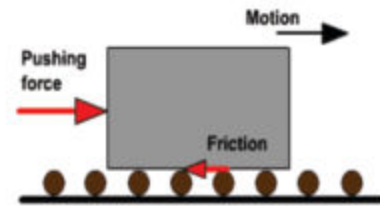


Figure 6.7 ball bearings as, used to roll the object.

Air resistance

- Explain air resistance.
- Identify the ways to reduce air resistance

When you ride a bike or run, you feel that air is pushing you back.

Moving objects like aircrafts, cars and arrows experience similar air resistance when they are in motion. Frictional forces between the air and the moving object cause this resistance. There is more resistance with faster movement and less resistance with slower movement.



Figure 6.8 Cycling

Air resistance is a force (friction) that opposes any object moving through the air.



Figure 6.9: streamlining the body to reduce air resistance



This boy is bending himself to reduce the air resistance and making his body **streamlined**.

He is wearing smooth clothing and helmet designed to overcome air resistance. This makes him glide through the air with faster speed.



Cars, aeroplanes and many fast moving objects are usually streamlined to overcome air resistance.



Figure 6.10: streamlined car and aeroplane

Swimmers make their bodies streamlined to reduce friction under water. This helps them swim easily and smoothly through water.



Figure 6.11: Swimmers

On the other hand parachutes increase air resistance due to the spread in surface area and allows the sky diver to easily descend towards the Earth.

Gravity

 **Explain**
gravity as a force.

Do you know?

Gravity also holds Earth and the other planets in their orbits around the Sun.

Do you think these two things when dropped would hit the ground at the same time?



Gravity is a pull or force of attraction between two objects and is a property of all matter.

The more gravity of earth is for heavy mass and an object has, the bigger its attraction is to another object.

The Earth is a huge object with a lot of mass. Everything on earth is pulled towards its center. Our weight is caused by the pull of gravity. Our weight is a force with which the earth attracts us towards its center.

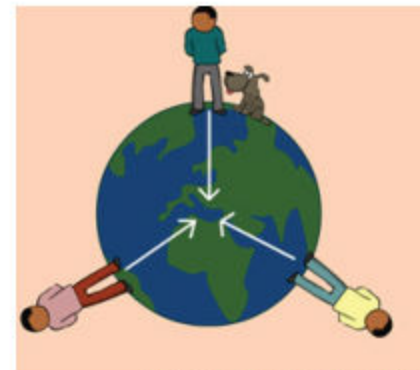


Figure 6.12: Earth pulling the objects towards its center

Mass And Weight

Which of the following has more mass? A bowling ball or a balloon?

 **Differentiate**
between mass and weight.



Figure 6.13: Bowling ball and balloon

Every object is made up of matter. **Mass is the quantity of matter in an object.** You are often confused between mass and weight.

Weight is not the same as mass. Weight is the pull of gravity that acts on the mass of an object. Mass is measured in **kilogram (kg)** while weight is measured in newtons (N). Mass is independent of gravity. It means that mass remains the same on Earth or on any other heavenly body, but weight changes due to the difference in gravity.

Thus if you were to travel to the Moon your weight would change because the pull of gravity is weaker there than on Earth but, your mass would stay the same because you are made up of same amount of matter. This is shown in the figure on your right.

Weight = mass $\times$ gravity.

The gravity on moon is $\frac{1}{6}$ of the Earth.

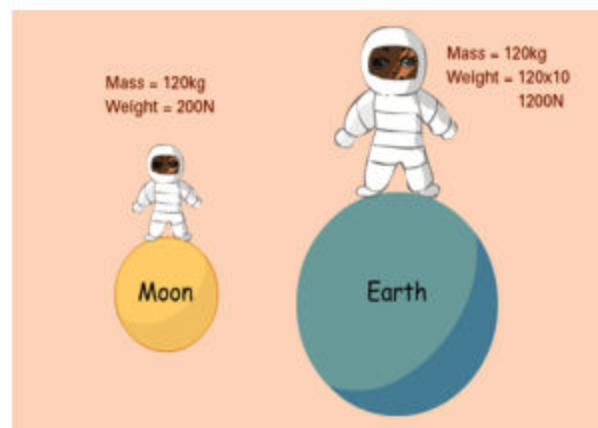


Figure 6.14: The Earth and the Moon

Activity 2: Identify the difference between mass and weight

Things I need:

- A school bag
- All your school books

What to do:

1. Take out your entire books and place it on the table. Now, try lifting the school bag with two of your fingers (middle and index fingers).
2. Next, add a few books and try to lift it.
3. Keep on adding your books to the bag, and try lifting it.
4. Repeat step 2, until all books are inside the bag, now lift the bag.

Activity questions:

1. Was it harder to lift the bag without the books or the full bag? Why?
2. Why do you think that adding books to the bag made it more difficult to lift?
3. What happened to the mass and weight of the bag, as more books were added to it?

What I conclude:

- 1 Differentiate between balanced and unbalanced forces.
- 2 Describe the effects of balanced and unbalanced forces on the motion of an object.

Balanced and unbalanced forces

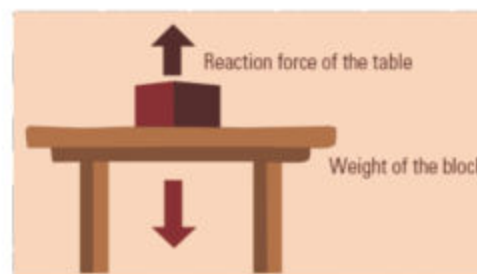
These people are engaged in a tug of war. Both the teams are applying forces but none of them are moving. Can you think why?



Figure 6.15: Tug of war

When two equal sized forces are applied on an object in opposite directions, the object does not move. We say that the forces are balanced. Anytime when an object experiences balanced forces, it stays still, or continues to move in the same speed and in the same direction.

In the figure on your right, a block of wood is resting on the table. The block is not moving because two forces are acting on it. The forces are the weight of the block, which acts downwards and reaction force of the table which acts in upward direction.



Balanced forces can be demonstrated through the hanging, floating and sitting objects.



Figure 6.16: Balanced forces

When forces are unbalanced, there is motion:



Figure 6.17: Unbalanced forces

Imagine you are riding a bike on a flat road. When forward forces are bigger than opposing forces you accelerate. When the forces are balanced, your speed stays the same.



Inertia

 Explain inertia.



Imagine you are sitting in a bus, when the bus starts moving, you are likely to fall backward. But when the bus stops moving, you are likely to fall forward. Do you know why this happens? This happens because when the bus moves suddenly from the state of rest, our feet are carried forward along with the moving bus but the inertia of our body tends to keep us at rest. This causes our body to fall backwards. When the bus stops suddenly, our feet are brought to rest with the bus, but the inertia of our body tends to continue its forward motion. This causes our body to fall forward.

The inertia of an object is the tendency of the object to remain at rest or, if moving, to continue its uniform motion in a straight line.”

Activity 3: Demonstrating Inertia

What I need:

- a card piece
- a glass
- a coin



What I do:

1. Place the card piece on an empty glass.
2. Put a coin on the card piece.
3. Flick the card from the glass with a sharp blow of your finger. Record your observation.



What I observed:

When the card is flicked.....

Activity questions:

1. What happens when the card is flicked?
2. Why is it necessary to flick the card with a sharp blow?
3. What would you conclude about above activity?

Levers

- Describe simple machines (wedge, inclined plane, and kinds of levers)
- Describe the use of wedge, inclined plane and levers in daily life.

Look at the following figures and select the object that you would choose to open a can.



a spoon



stones



a tin can

In the figure above, spoon acts as a lever.

Teacher Note: This activity can be done individually. Ask the students to bring a glass and a coin. Engage the students in the activity and provide them an explanation.

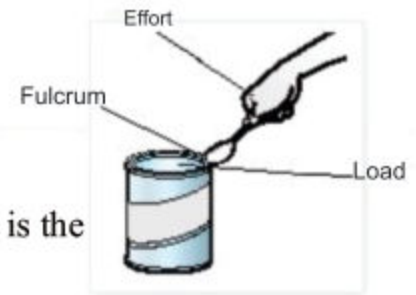
A **lever** is a long bar or stick which moves around a fixed point called **fulcrum**.

Lever is a kind of a **simple machine**. Simple machines make our work easier by reducing the amount of effort.

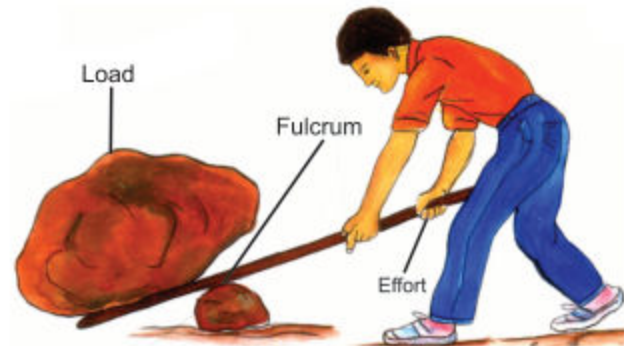
In the given figure, the spoon would act as a lever.

The lid which you want to remove is a **load**.

The fixed point (corner of the can) on which the spoon moves is the **fulcrum**. The force which you apply to open the load is the **effort**.



Similarly this man is trying to move a large stone with the help of a long rod which acts as a lever.



Do you know? Levers have been used since the prehistoric times for cultivation, excavation, and moving large objects. The oldest archaeological findings of a lever dates back to the Archimedes, Greek mathematician and physicist during the 3rd century BC. The Egyptians used the lever to move things over 100 tons.



Figure 6.18: Using lever to move heavy load

Kinds of levers

According to the positions of effort, a fulcrum and load, lever is divided into three kinds.

First kind of lever

In the first kind of lever, the fulcrum is between the effort and the load.

First kind of lever

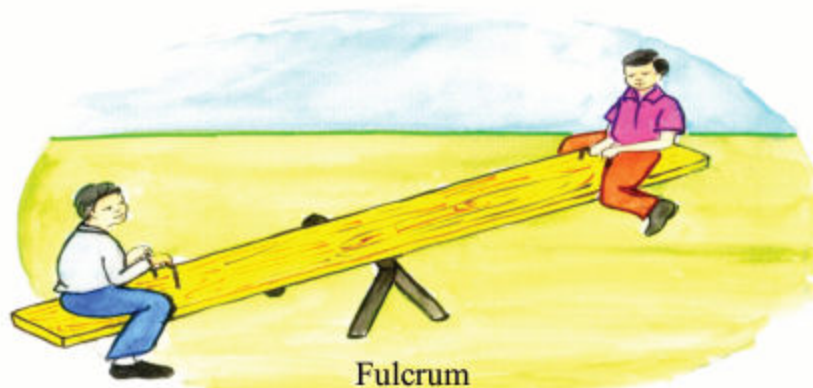


Figure 6.19: Lever in a see-saw

Second kind of lever:

In the second kind of lever, the load is between the effort and the fulcrum.



Figure 6.20: Lever in a wheel-barrow

Third kind of levers:

In the third kind of lever, the effort is between the load and fulcrum.

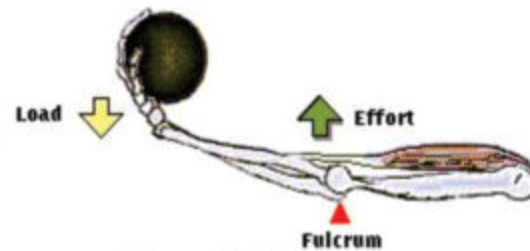


Figure 6.21: Lever in a an arm

Activity 4: Identifying the three kinds of levers

What do I need:

- a nail cutter
- a stapler
- a bottle opener
- a pair of pliers
- a pair of scissors
- a pair of tongs
- Pictures of a person brooming and wheel barrow

What I do:

1. Identify the positions of fulcrum, effort and load in each of the item mentioned above.
2. Classify each item as first, second and third kind of levers and record your observations in the table.

[Note: The students will perform this activity in groups.]

What I observed:

First kind of lever	Second kind of lever	Third kind of lever

Activity questions:

1. Which items are the first kind of levers?
2. Which items are the second kind of levers?
3. Which items are the third kind of levers?
4. Which of the kinds are the easiest to identify? Why?

Levers around us

Levers make our lives easier. In your home you may find different kinds of levers.

Some are shown below:



a nut cracker



a bat and ball

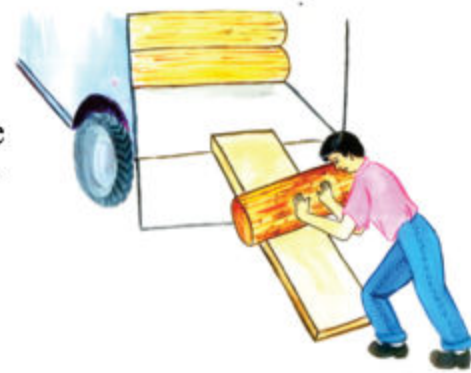
- a pair of scissors
- a pair of tongs
- a see-saw
- a pair of plier



A cort, what is this called?

Inclined plane:

This man is rolling a heavy drum on the wooden ramp. Why is he doing it? Will he be able to move the drum up without the ramp? A ramp is an example of an inclined plane. An **inclined plane** is a type of a simple machine. It is a flat and sloppy surface used to move heavy objects up or down over a vertical height.



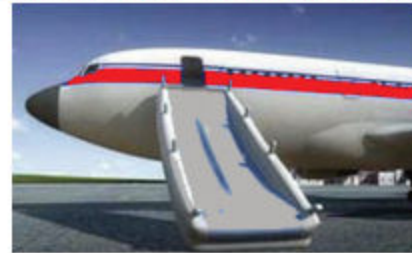
Inclined plane around us



Ramps for wheelchairs



Ramps are used for loading and unloading



In plane emergency exit



Ramps for moving objects

Figure 6.22: Different inclined planes

Activity 5: Demonstrating an inclined plane

What I need :

- 2-4 thick books
- a large stone or potato
- a string
- a small board or heavy cardboard for the inclined plane
- a spring scale

What I do :

1. Stack the books on the table top or the floor.
2. Place one end of the cardboard on the stack of the books and the other end on the table (or the floor).
3. Tie the string around the stone. Make a loop at the other end of the string and hang it on the hook of the spring balance.
4. Lift the stone from the table top (or floor) to the top of the stack of books (vertically) with the help of the spring balance and note down the force in newtons and record it in the table.
5. Now pull the stone up the inclined plane from the table top (or floor) towards the top of the books. Note down the reading on the spring scale and record it in the table.



What I observed:

Force (N) applied to lift the stone without the inclined plane	Force (N) applied to lift the stone using the inclined plane

Activity questions:

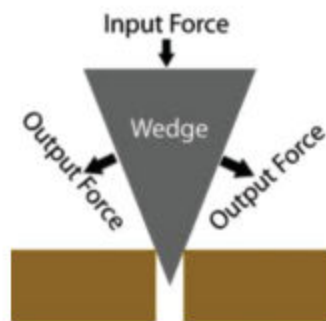
1. How much force is applied to lift the stone without using the inclined plane?
2. How much force is applied to lift the stone using the inclined plane?
3. What is the difference of force applied in both the cases?
4. Experiment with different heights by adding or removing books under the inclined plane. Is it more difficult to move the stone over a steep incline, a short horizontal distance or a gradual incline over a longer horizontal distance?

Wedge

A wedge is a type of a simple machine with two slanting sides ending in a sharp edge, which cuts the material apart. A wedge has two inclined planes which join back to back. A wedge is used to cut things. This man in the figure right, is cutting the wood with an axe which has a wedge shaped blade. He pushes the wood downward with force and the wedge shaped blade pushes the wood apart.



Figure 6.23: Man cutting wood with a wedge shaped axe



Do you know the types of wedges around you?



cleaver



cook's knife



carving knife



+ bread knife

Kitchen knives



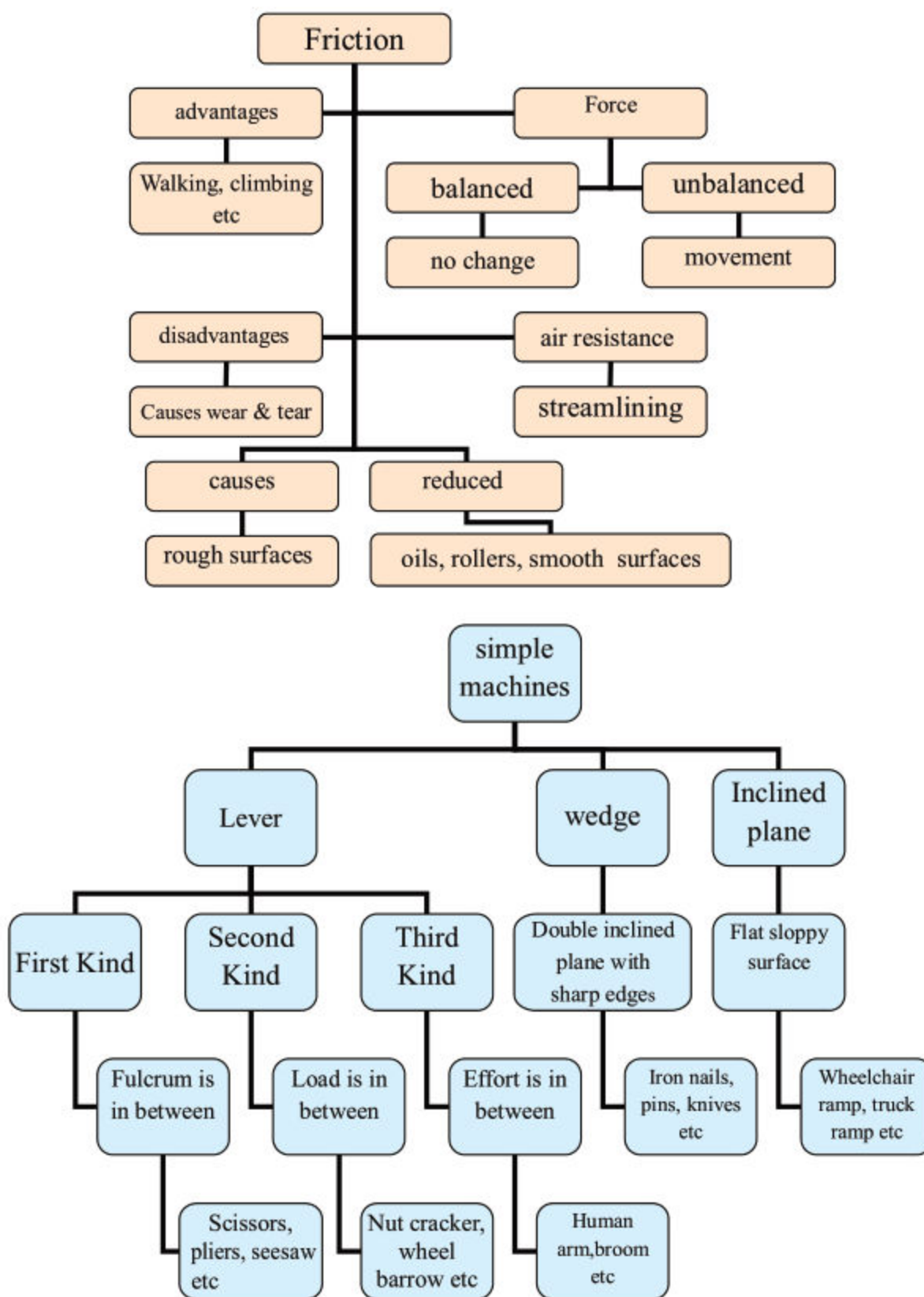
Push pin

Iron nails



Figure 6.24: Different types of wedges

Summary



Review questions :

1. Locate fulcrum, effort and load in each of the following items by writing F,E and L respectively in the figures.



2. Fill in the blanks.

- a) A stapler is _____ kind of lever.
- b) _____ is a force you put into the simple machine.
- c) _____ is the fixed point around which a lever moves.
- d) Unbalanced forces cause _____ in an object.
- e) _____ is an example of a wedge.

3. Give two reasons for the following statements.

- a) Car tyres have treads.
- b) Air crafts have streamlined bodies.

4. Explain with the help of one example from daily life, why frictional forces depend on the

- a) surfaces in contact
- b) weight of the object

5. What are unbalanced forces? Explain with the help of one example.

6. Tick the correct answer.

- i. Which of the following is the second kind of lever?
 - a) A nail cutter
 - b) A fishing rod
 - c) Pliers
 - d) Claw hammer
- ii. Which of the following can be used to reduce friction between the moving parts of machines?
 - a) sand
 - b) grease
 - c) gears
 - d) screws
- iii. Unit of mass is
 - a) Newton
 - b) kilogram
 - c) joules
 - d) centimetre

Inquiry project:

Find different types of simple machines around your school and home.

Draw or stick a picture of a simple machine.	State the function